GREAT SALT LAKE DUST MODEL

Implementation Plan: The Stages That Bring the Model to Life

*A stage-by-stage plan built entirely on **GSL_Dust_Model_Guide.pdf**, the project's modeling standard. Every stage follows the guide's build order, every method follows the guide's component table, and every assumption the guide asks us to test is assigned to a stage where it gets tested.*

Prepared 30 September 2026 · Items marked VERIFIED were checked against the live source on that date
Supersedes the earlier "V2 Implementation Plan" · Standard: docs/GSL_Dust_Model_Guide.pdf

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1. How this plan relates to the guide

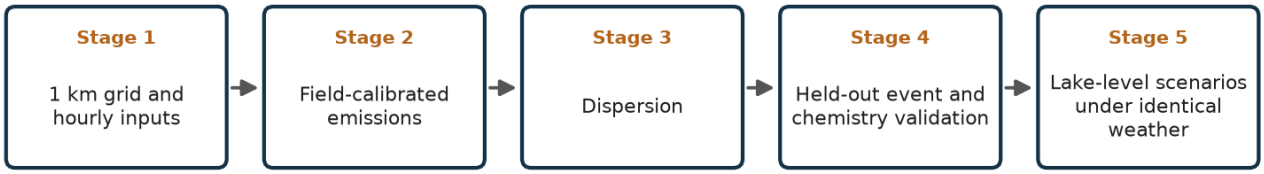
The guide is the specification: it fixes *what* the model is (the equations, the compartments, the outputs and where the model stops), *which method* each component uses, *which assumptions* must be tested, and *the order* the work happens in. This plan adds only the *how*: where each input lives, what it takes to get it into the model, what "done" looks like at each stage, and where the available data falls short of what the guide's method needs.

Three rules follow from that:

1. **The five stages are the guide's build order**, in the guide's sequence, with no stages added or merged.
2. **Methods are the guide's methods.** Where the data can't yet support a guide method, the plan reports a **data-readiness gap** against that method; it does not quietly swap in a different method. The only alternatives used are the ones the guide itself names ("otherwise physics" for φ; "measured or bounded" for η; "particle bins or uncertainty bounds" for size effects).
3. **Fidelity is checkable.** Appendix A maps every requirement in the guide to the section that satisfies it.

2. The standard, restated

> **GOAL AND SCOPE (GUIDE)**
> Predict PM10 dust and its arsenic moving from Great Salt Lake sediment into the Salt Lake Valley. Outputs are **arrival mass** and **arrival rate**; the model **stops before health effects**.



The guide's build order. Each box is one stage of this plan.

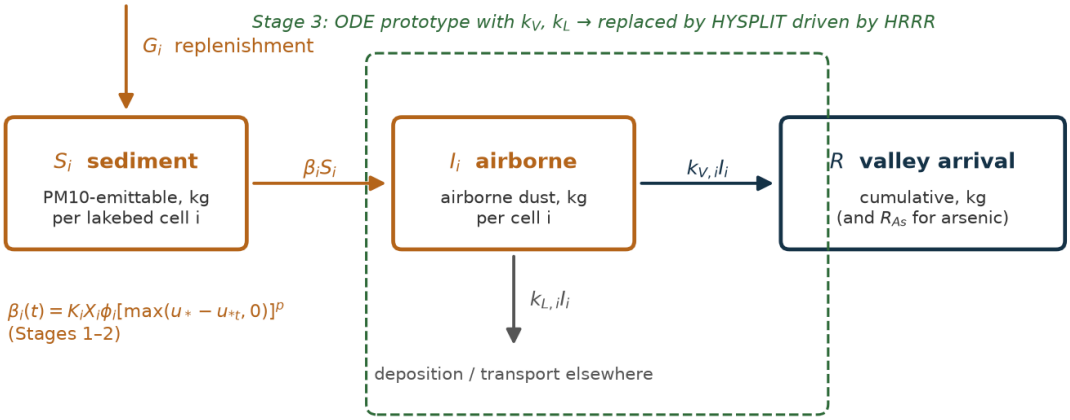
$$\beta_i(t) = K_i\, X_i(t)\, \phi_i(t)\, [\max(u_{*,i}(t) - u_{*t,i}(t),\ 0)]^p$$

$$\frac{dS_i}{dt} = G_i - \beta_i S_i \qquad \frac{dI_i}{dt} = \beta_i S_i - (k_{V,i} + k_{L,i}) I_i$$

$$\frac{dR}{dt} = \sum_i k_{V,i}\, I_i \qquad \frac{dR_{As}}{dt} = \sum_i c_{As,i}\, \eta_i\, k_{V,i}\, I_i$$

The governing equations, exactly as specified in the guide.

SYMBOL	MEANING (GUIDE)	UNITS / RANGE
`S_i`	PM10-emittable sediment in lakebed cell i	kg
`I_i`	Airborne dust from cell i	kg
`R`, `R_As`	Cumulative dust (and arsenic) entering the valley	kg; rates dR/dt, dR_{As}/dt in kg per time
`X_i(t)`	Exposed fraction of the cell, from lake elevation and terrain	0–1
`φ_i(t)`	Erodibility	0–1
`u*`, `u*t`	Friction velocity and its erosion threshold	m/s
`p`	Wind sensitivity	dimensionless
`K_i`	Emission coefficient, calibrated so β has inverse-time units	1 / (time · $(m/s)^p$)
`G_i`	Sediment replenishment	kg per time
`k_V,i`, `k_L,i`	Transfer into the valley; removal by deposition or transport elsewhere	1 / time
`c_As,i`, `η_i`	kg arsenic per kg sediment; enrichment in airborne particles	kg/kg; dimensionless



Mass flows from sediment to airborne dust to valley arrival. The dashed region is the part the guide says to prototype with first-order transfer, then replace with HYSPLIT dispersion driven by HRRR.

3. Stage 1 — 1 km grid and hourly inputs

Builds the spatial frame (lakebed cells i), the exposure term $X_i(t)$, and every hourly forcing the later stages need. Guide method for exposure: **geometry; no ML**.

1.1 Build the 1 km lakebed grid

- Cells i cover the lakebed, since the ODE is written "for each lakebed cell i". Use a projected equal-distance grid (UTM zone 12N, NAD83, matching the terrain data's horizontal datum) so every cell is exactly 1 km × 1 km.
- Tag each cell with its arm/bay (north arm, south arm, Farmington Bay, Bear River Bay), because lake level is applied by arm.
- The Salt Lake Valley boundary is needed as a receptor (it defines "entering the valley" for R and for the first-entry rule in Stage 3). The guide does not draw it, so it's listed under Decisions.

1.2 Exposure $X_i(t)$ — geometry, no ML

Inputs (guide): USGS Saltair lake gauge; USGS terrain / Utah lidar.

- **Lake level by arm.** Saltair (USGS 10010000) for the south arm and Great Salt Lake near Saline (USGS 10010100) for the north arm, both daily VERIFIED.
- **Terrain.** The USGS half-meter topobathymetric elevation model (bathymetry 2002–2006 merged with 2016 Utah lidar) VERIFIED. Horizontal NAD83, vertical **NAVD88, meters**.
- **Method.** For each cell, $X_i(t)$ = fraction of the cell's area whose lakebed elevation is above that arm's water level at time t.

  Computed from the terrain pixels inside the cell, so exposure isn't reduced to one average elevation per cell.

> **GUIDE ASSUMPTION: MATCH ELEVATION UNITS AND VERTICAL DATUMS**
> This is a live issue, not a formality. The gauges report parameter 62614, which is **NGVD 1929, feet** VERIFIED; the terrain is **NAVD 1988, meters**. USGS's own conversion at the Saltair gauge (NOAA VERTCON) is **NAVD88 = NGVD29 + 3.47586 ft** VERIFIED. Stage 1 converts every gauge reading to NAVD88 meters before it touches the terrain, with a unit test on the conversion. (The earlier v1 code labelled these same gauge readings as NAVD88, which is exactly the error this assumption exists to catch.)

Terrain access (implementation detail)

The terrain is a single 34.0 GB GeoTIFF plus a 9.8 GB overview file on ScienceBase, stored behind tokenized S3 download links (the ScienceBase API / `sciencebasepy` issues them) VERIFIED. A 1 km cell doesn't need half-meter pixels: USGS itself resampled to 5 m to

build its area tables. Plan: read a coarse overview level (about 5–25 m) rather than the full-resolution file, and record the resolution used, since the guide asks for a grid-resolution test.

Built-in validation target

USGS publishes elevation-area tables for the total lake and each of the four basins at 0.01 ft steps from 4,170 to 4,215 ft NAVD88, computed from this same terrain VERIFIED. Summing the wet area of each basin's cells at each level must reproduce those tables. That's a direct check on the grid, the basin tags, and the datum handling together.

1.3 Hourly inputs

INPUT	SOURCE	USED BY
Friction velocity $u^*_i(t)$	NOAA HRRR surface field `FRICV` VERIFIED, checked against local weather stations	$\beta_i(t)$ threshold law (Stage 2)
Winds and mixing	NOAA HRRR (10 m winds, PBL height `HPBL` VERIFIED); local weather stations	Transport (Stage 3)
Moisture / surface state	Precipitation and humidity (HRRR, stations); satellite imagery	$\varphi_i(t)$ (Stage 2)
Lake level by arm	USGS 10010000, 10010100 (daily → held constant within each day)	$X_i(t)$

GUIDE ASSUMPTIONS TESTED IN STAGE 1

Match elevation units and vertical datums (conversion + unit test) · **use lake levels by arm** (two gauges, basin tags) · **test grid resolution** (recompute X at 1 km and at a finer and a coarser grid, and report how much total exposed area and its spatial pattern change) · **cells use average properties** (X is computed from sub-cell terrain; the cell-average assumption is carried into φ in Stage 2).

STAGE 1 IS DONE WHEN

the grid and basin tags exist; per-basin wet area reproduces the USGS elevation-area tables (target within 1–2%, since both come from the same terrain); datum conversion is unit-tested; the grid-resolution comparison is recorded; and hourly $X_i(t)$, $u^*_i(t)$ and moisture inputs are complete for the chosen simulation period.

4. Stage 2 — Field-calibrated emissions

Turns exposure and wind into emission $\beta_i(t)$ and tracks the sediment reservoir S_i. The guide groups **emission and chemistry** together, so the arsenic fields $c_{As,i}$ and η_i are also built here, ready for Stage 3.

2.1 Erodibility $\phi_i(t)$

Inputs (guide): PI-SWERL field tests; moisture, crust, texture, satellite imagery. **Method (guide):** RF/XGBoost for φ with field labels; otherwise physics.

- With field labels: PI-SWERL measurements become training labels; moisture, crust, texture and satellite features are the predictors; evaluation holds out whole field sites, not random samples.
- Without them: physics-based φ from the same moisture/crust/texture inputs (the guide's "otherwise physics").
- φ must be able to reach 0: the guide notes exposed ground may be crusted and non-erodible.

2.2 Threshold law: K_i, $u^*_{t,i}$, p

- The guide calls the law approximate and says to **calibrate it against field fluxes**. PI-SWERL measures emission flux at controlled friction velocities, which is exactly the data that pins down u^*_t, p and K.

- Calibrate K so β comes out in inverse-time units (guide).
- K, p and u^*_t are calibrated against field fluxes, not tuned to PM10 monitors. That keeps emission independent of the Stage 4 validation data.

2.3 Sediment reservoir: $S_i(0)$ and G_i

- The guide states initial S and replenishment G **need independent estimates**. Candidate independent evidence: supply-limited (depleting) behavior in repeated field flux tests, and surface-sediment surveys. Carry both as ranges into the uncertainty budget.
- Integration detail: with coefficients held constant within each hour, $dS/dt = G - \beta S$ has an exact exponential solution per step, which stays stable when β is large.

2.4 Arsenic: $c_{As,i}$ and η_i

Inputs (guide): USGS sediment and dust chemistry; particle-size-matched samples. **Method (guide):** GP/kriging for c_{As}; measured or bounded η.

The dataset the guide links (USGS, "Dust and sediment data from Great Salt Lake and northern Utah, 2022") was checked VERIFIED:

WHAT IT CONTAINS	IMPLICATION FOR THE GUIDE'S METHOD
71 samples: 34 passive dust-trap (150 cm), 14 surface sediment, 17 water, 2 standards. Arsenic in ppm (= mg/kg, so 10^{-6} kg/kg) for 17 dust and 14 sediment samples.	Directly usable c_{As} observations; ppm converts to the guide's kg/kg.
Only three sediment samples are at playa/lakeshore sites ("Playa west of OBW bundt site" 13.7 ppm, "The Boneyard" playa 13.2 ppm, "Promontory Road" 15.1 ppm). The other eleven are urban, farm and construction sites (0.9–3.3 ppm).	Three lakebed points cannot support a meaningful GP/kriging surface over the lakebed. The non-lake samples are still useful as contrasting source signatures.
All samples are the <63 µm size fraction.	Not PM10-size-matched, so η can't be measured from this release; it must be bounded until size-matched samples exist.

DATA-READINESS GAP (CHEMISTRY)

The guide's GP/kriging method is kept. What's missing is lakebed sample density. Before kriging, gather additional lakebed sediment chemistry (other USGS/state/university releases, or new sampling), then report kriging variance as the chemistry uncertainty. Size-dependent transport and enrichment use the guide's options: particle bins or uncertainty bounds (see Decisions).

DATA-READINESS GAP (FIELD CALIBRATION)

No public PI-SWERL dataset for Great Salt Lake was located during this plan's source check. PI-SWERL results are the gating input for both the RF/XGBoost φ model and the threshold-law calibration, so locating published field campaigns (or requesting their data) is the first sourcing task of Stage 2.

STAGE 2 IS DONE WHEN

the threshold law is calibrated against field fluxes with parameter uncertainty; φ is evaluated on held-out field sites (or the physics φ is documented); $S_i(0)$ and G_i have independent estimates with ranges; and $c_{As,i}$ (kriged, with variance) and η (measured or bounded) exist for every lakebed cell.

5. Stage 3 — Dispersion

Moves emitted dust from the lakebed to the valley. The guide specifies two steps: **prototype with the ODE**, then **replace its airborne transfer equations with HYSPLIT dispersion driven by HRRR**, using emissions βS. Guide method: HYSPLIT; no ML.

3.1 ODE prototype (first-order transfer)

- Implement dI/dt, dR/dt and dR_{As}/dt with per-cell $k_{V,i}$ and $k_{L,i}$, under the guide's stated assumption of well-mixed compartments and first-order transfer.
- Purpose: prove the kg bookkeeping end to end. Checks: mass closes to numerical precision (sediment lost = airborne gained + valley arrival + removed); R never decreases; first valley entry is counted once (in the ODE, dust moved into R leaves I, so this holds by construction).
- The prototype's k_V, k_L need a documented physical basis (for example cell-to-valley distance and how often HRRR winds blow toward the valley). They aren't tuned to PM10: R is cumulative arrival, not concentration.

3.2 Replace transfer with HYSPLIT driven by HRRR

- Source term: $\beta_i S_i$ per cell per hour. HYSPLIT then handles transport and deposition in place of k_V and k_L, and provides concentrations for monitor comparisons (guide).
- HYSPLIT is public from NOAA READY; HRRR 3 km is available pre-formatted for HYSPLIT on a public AWS bucket, June 2015 onward VERIFIED.
- **Count first valley entry once.** With particle output, flag each particle's first crossing into the valley boundary and add its mass to R (and R_{As}) only then, so dust that leaves and re-enters isn't double counted.
- Run HYSPLIT and the ODE prototype over the same period and compare where their arrivals differ; the differences show what first-order transfer was missing.

> **ENGINEERING CONSTRAINT: HRRR DATA VOLUME**
> The HYSPLIT-ready HRRR files cover the whole continental US at about 3.4 GB per 6-hour file, roughly 14 GB per simulated day VERIFIED, and a spatial subset can't be pulled from them remotely. Two workable paths, still "HYSPLIT driven by HRRR": (a) run event windows from that archive, or (b) build a regional Utah file from the original HRRR output, which can be downloaded variable by variable and cropped, then converted with HYSPLIT's own conversion tools. Choose based on how many simulated days Stages 4–5 need.

> **GUIDE ASSUMPTIONS TESTED IN STAGE 3**
> Well-mixed compartments and first-order transfer (tested by comparing the ODE prototype against HYSPLIT) · count first valley entry once · R is cumulative arrival, not airborne concentration.

> **STAGE 3 IS DONE WHEN**
> the ODE prototype closes its mass balance; HYSPLIT runs from the hourly βS source; the first-entry rule is implemented and tested on a case with re-entering particles; and hourly concentrations exist at every validation monitor, along with R(t) and $R_{As}(t)$.

6. Stage 4 — Held-out event and chemistry validation

Inputs (guide): EPA AQS / Utah DAQ PM10; independent plume and filter chemistry evidence. **Method (guide):** held-out events and locations.

- **Hold out both.** Fix, before any comparison, a set of whole dust events and a set of whole monitor locations that are never used for calibration. Score the model on held-out events at held-out locations as well as each separately.
- **Compare like with like.** PM10 monitors are compared with HYSPLIT concentrations at the monitor, never with R.

- **Attribution needs more than PM10.** The guide states PM10 alone cannot identify GSL dust. Each validated event needs independent plume evidence that the dust came from the lakebed.
- **Arsenic needs chemistry.** PM10 cannot validate arsenic (guide). Modeled arsenic is checked against filter chemistry. The USGS 2022 release's dust-trap arsenic (17 samples) is one independent line of evidence; it measures deposition over a collection period, so the comparison is against modeled deposition over the same period, not against concentration.

> **STAGE 4 IS DONE WHEN**
> held-out sets are fixed in advance; PM10 skill is reported for held-out events and locations; each validated event has independent plume evidence; and modeled arsenic is compared against independent chemistry with the chemistry uncertainty from Stage 2.

7. Stage 5 — Lake-level scenarios under identical weather

- Hold the hourly HRRR weather fixed and change only lake level (by arm). The change flows through $X_i \rightarrow \beta_i \rightarrow$ emission $\rightarrow$ HYSPLIT $\rightarrow$ R and R_{As}.
- Report arrival mass (R, R_{As}) and arrival rate (dR/dt, dR_{As}/dt) per scenario, each with its uncertainty range.
- Scenario outputs stop at arrival. No exposure or health-effect estimates, as the guide specifies.
- Scenario lake levels are listed under Decisions.

8. Uncertainty reporting (all stages)

The guide asks for uncertainty ranges for **emission, chemistry and transport**, reported separately so it's clear which one dominates.

COMPONENT	SOURCES OF UNCERTAINTY	HOW IT'S CARRIED FORWARD
Emission	φ model error on held-out field sites; threshold-law parameters (K, u^*_t, p); ranges on $S_i(0)$ and G_i; grid-resolution sensitivity	Ensemble of emission runs sampling these ranges
Chemistry	Kriging variance of c_{As}; bounds on η; particle-size treatment	Sampled alongside emission for R_{As}
Transport	HYSPLIT configuration and HRRR error; the ODE-versus-HYSPLIT structural difference	Perturbed or alternate transport runs

9. Data readiness

SOURCE (AS LINKED IN THE GUIDE)	ROLE	STATUS
USGS Saltair gauge 10010000	Exposure X (south arm)	VERIFIED Daily, NGVD29 ft
USGS gauge 10010100 (near Saline)	Exposure X (north arm)	VERIFIED Daily, NGVD29 ft. Not linked in the guide, but needed for "lake levels by arm"
USGS topobathymetric model and area tables	Exposure X	VERIFIED NAVD88 m; 34 GB + 9.8 GB overviews, tokenized access; per-basin tables
Utah lidar (UGRC)	Exposure X	2016 GSL lidar is already merged into the topobathymetric model
PI-SWERL field tests	φ, threshold law	**Not yet located** — gating item for Stage 2
USGS dust and sediment chemistry, 2022	c_{As}, η	VERIFIED Downloadable; 3 lakebed sediment points; <63 µm only
Particle-size-matched samples	η	**Not yet located** — η bounded until found

NOAA HRRR (HYSPLIT-ready archive)	u*, transport	VERIFIED Public AWS, June 2015 onward; ~14 GB/day
HYSPLIT (NOAA READY)	Transport	VERIFIED Public
EPA AQS / Utah DAQ PM10	Validation	Public (EPA bulk hourly files; Utah DAQ portal)
Independent plume evidence; filter chemistry	Validation	**To be sourced**

10. Decisions needed before building

The guide leaves these open. Each needs a choice (yours) before the stage that uses it:

1. **Salt Lake Valley boundary** (Stage 1, used in Stage 3). Defines "entering the valley" for R and the first-entry rule.
2. **Lake-level source for Farmington Bay and Bear River Bay** (Stage 1). The two USGS gauges cover the north and south arms; these bays are partly separated by causeways and dikes.
3. **Particle bins or uncertainty bounds** for size-dependent transport and enrichment (Stage 2–3). The guide allows either.
4. **Sourcing path for field data** (Stage 2): published PI-SWERL campaigns, data requests, or new fieldwork; same for more lakebed chemistry.
5. **Simulation period and HRRR strategy** (Stage 3): event windows from the HYSPLIT-ready archive, or a regional file built from raw HRRR.
6. **Scenario lake levels** (Stage 5).

Appendix A — Guide requirement traceability

REQUIREMENT IN THE GUIDE	WHERE THIS PLAN MEETS IT
Predict PM10 dust and its arsenic moving from GSL sediment into the Salt Lake Valley	Stages 1–5
Outputs are arrival mass and arrival rate	3.1, 3.2, Stage 5
The model stops before health effects	Stage 5
S, I, R are masses in kg per lakebed cell i	§2 symbol table; 3.1
X is the exposed fraction, from lake elevation and terrain	1.2
φ erodibility (0–1)	2.1
u* friction velocity, u*t erosion threshold	1.3, 2.2
G replenishes sediment	2.3
kV transfers into the valley; kL removes by deposition or transport elsewhere	3.1
β and k have inverse-time units; calibrate K accordingly	2.2
p controls wind sensitivity	2.2
cAs kg/kg; η enrichment in airborne particles	2.4
Prototype with the ODE	3.1
Replace airborne transfer with HYSPLIT driven by HRRR, using emissions βS	3.2
HYSPLIT handles transport and deposition and provides concentrations for monitor comparisons	3.2, Stage 4
Exposure X: gauge + terrain/lidar; geometry, no ML	1.2
Erodibility φ: PI-SWERL, moisture, crust, texture, imagery; RF/XGBoost with field labels, otherwise physics	2.1
Arsenic: USGS chemistry, size-matched samples; GP/kriging for cAs; measured or bounded η	2.4
Transport: HRRR winds and mixing, local stations; HYSPLIT, no ML	1.3, 3.2
Validation: AQS / Utah DAQ PM10, independent plume and filter chemistry; held-out events and locations	Stage 4
Cells use average properties; exposed ground may be crusted and non-erodible	1.2, 2.1
Match elevation units and vertical datums	1.2
Use lake levels by arm	1.1, 1.2, Decision 2
Test grid resolution	Stage 1 assumptions box; §8
Initial S and replenishment G need independent estimates	2.3
Calibrate the approximate threshold law against field fluxes	2.2
Arsenic rides on dust with a cell-specific composition	2.4 ($c_{As,i}$ per cell)
Size-dependent transport and enrichment: particle bins or uncertainty bounds	2.4, Decision 3
ODE assumes well-mixed compartments and first-order transfer	3.1, 3.2 comparison
Count first valley entry once	3.1, 3.2
R is cumulative arrival, not airborne concentration	3.1, Stage 4

PM10 alone cannot identify GSL dust or validate arsenic	Stage 4
Build order: grid → emissions → dispersion → validation → scenarios	Stages 1–5
Report uncertainty ranges for emission, chemistry and transport	§8

Appendix B — What changed from the earlier V2 plan

- **Structure:** nine self-defined phases replaced by the guide's five-stage build order.
- **Monitor comparison:** the earlier plan proposed converting R into a concentration with an assumed mixing volume. Removed: the guide says R is not concentration, and monitor comparisons use HYSPLIT concentrations.
- **Arsenic:** the earlier plan led with literature placeholder values. Now the guide's GP/kriging method stands, with the lack of lakebed samples reported as a data gap; η is bounded only as the guide allows.
- **Erodibility:** physics φ is no longer presented as the default; it's the guide's fallback when field labels are unavailable.
- **Design drivers:** the earlier plan was shaped around reusing v1 code. This plan is shaped only by the guide; code reuse is an implementation convenience, not a design input.
- **New, verified facts added:** gauge datum (NGVD29), terrain datum (NAVD88 m), the official offset, north-arm gauge, terrain file sizes and access, contents of the guide's chemistry dataset, HRRR friction velocity and data volume.

Sources are those linked in GSL_Dust_Model_Guide.pdf: USGS monitoring location 10010000; USGS half-meter topobathymetric elevation model and elevation-area-volume tables; UGRC SGID lidar; USGS dust and sediment data from Great Salt Lake and northern Utah, 2022; NOAA ARL HYSPLIT meteorological archives; NOAA READY HYSPLIT; EPA AQS.